Vector spaces and bases

Original CC0 evaluation fixture - generated locally

A vector space is a set with vector addition and scalar multiplication satisfying the ve ctor space axioms.

A basis is a linearly independent spanning set. Every vector has a unique coordinate rep resentation in a chosen basis.

Linear maps

Original CC0 evaluation fixture - generated locally

A linear map T satisfies T(u+v)=T(u)+T(v) and T(av)=aT(v).

The rank-nullity theorem states dim(V) = rank(T) + nullity(T) for a linear map on a finite-dimensional domain V.